

ADVANCES IN HEART FAILURE, MECHANICAL CIRCULATORY SUPPORT AND TRANSPLANT

Next Generation Temporary Mechanical Circulatory Support for Cardiogenic Shock

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ABSTRACT: Cardiogenic shock remains the leading cause of death in patients with acute myocardial infarction and severe acute heart failure. Temporary mechanical circulatory support devices have emerged as important adjuncts for restoring systemic perfusion, unloading the failing ventricle, and stabilizing patients for cardiogenic shock and during high-risk percutaneous coronary interventions. However, currently available devices are associated with important limitations, including vascular complications, hemolysis, and uncertain survival benefit. Technological innovation has led to the development of new percutaneous mechanical support systems designed to deliver higher cardiac output with more limited vascular access and improved hemocompatibility. This review summarizes the current evidence regarding current and emerging temporary mechanical circulatory support devices used in cardiogenic shock and high-risk percutaneous coronary interventions. Literature from major clinical trials, observational registries, and device development studies was reviewed to evaluate device mechanisms, hemodynamic performance, and emerging technologies.

Key Words: assisted circulation ■ cardiac output ■ cause of death ■ heart-assist devices ■ heart failure ■ hemolysis ■ perfusion

Cardiogenic shock (CS) represents the most severe manifestation of acute heart failure and is characterized by inadequate cardiac output leading to systemic hypoperfusion and end-organ dysfunction.^{1,2} The syndrome is associated with high mortality, reaching 40% to 60% despite modern treatment strategies, particularly when it occurs in the setting of acute myocardial infarction (AMI).^{3–6} Acute on chronic and acute heart failure, as well as AMI, remain the most common causes of CS, accounting for ≈90% of cases.^{1,2} Other causes include secondary causes and postcardiotomy shock.⁷ Regardless of cause, CS is characterized by a vicious cycle of reduced cardiac output, impaired coronary perfusion, worsening myocardial ischemia, and progressive hemodynamic deterioration.

Temporary active mechanical circulatory support (tMCS) devices have become a key therapeutic strategy in patients with CS. In general, these devices aim to maintain coronary and systemic perfusion by augmenting cardiac output or by providing extracorporeal circulation. Depending on the device, they also unload the failing

left or right ventricle. Collectively, these mechanisms aim to interrupt the pathophysiologic shock spiral leading to multiorgan failure and finally death.

However, despite their increasing clinical use, evidence from randomized controlled trials (RCTs) remains limited. Up-to-date, only 1 trial investigating the routine use of a microaxial flow pump (mAFF) in a selected group of patients with ST-elevation myocardial infarction showed a reduction in mortality.³ In contrast, RCTs investigating extracorporeal life support (ECLS) in the venoarterial extracorporeal membrane oxygenation configuration (VA-ECMO) have not shown a survival benefit in CS.^{5,8} An individual patient data meta-analysis of all RCTs comparing tMCS versus control did not demonstrate an overall mortality benefit, although a signal toward improved outcomes was observed in patients with a low likelihood of hypoxic brain injury.⁹ On the contrary, all trials reported a substantial proportion of severe complications in the tMCS groups, including bleeding, limb ischemia, and specifically hemolysis and renal replacement therapy for mAFF devices. This underscores the

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Nonstandard Abbreviations and Acronyms

ALI	acute limb ischemia
AMI	acute myocardial infarction
CHIP-BCIS 3	Controlled Trial of High-Risk Coronary Intervention With Percutaneous Left Ventricular Unloading – British Cardiovascular Intervention Society 3
CS	cardiogenic shock
ECLS	extracorporeal life support
FDA	Federal Device Administration
HR-PCI	high-risk percutaneous coronary intervention
IABP	intraaortic balloon pump
LV	left ventricular
mAFP	microaxial flow pump
PULSE	Pulsecath Mechanical Support Evaluation
RCT	randomized controlled trial
RPM	revolutions per minute
SUPPORT	Early Feasibility Study of the Supira System in Patients Undergoing High-Risk Percutaneous Coronary Intervention
tMCS	temporary active mechanical circulatory support
UNLOAD-CHIP	Use of Mechanical Left Ventricular Unloading in Complex Higher-Risk Indicated Procedures
VA-ECMO	extracorporeal membrane oxygenation configuration

need for further development and optimization of tMCS devices.

In parallel, the use of tMCS has expanded to elective use in interventional cardiology. The increasing complexity of interventions has given rise to the concept of high-risk percutaneous coronary intervention (HR-PCI), where patients with advanced coronary artery disease and impaired left ventricular (LV) ejection fraction undergo complex revascularization procedures.¹⁰ tMCS in this setting aims at providing hemodynamic stability during these procedures, enabling complete revascularization in patients who otherwise might not tolerate prolonged ischemia.

Despite this strong physiological rationale, the use of prophylactic tMCS in HR-PCI has recently been challenged by randomized data as a multicenter RCT failed to demonstrate a clinical benefit of routine mAFP support.¹¹

Independently, an RCT in hemodynamically stable ST-elevation myocardial infarction patients also failed

to demonstrate an infarct size reduction and clinical benefit of routine mAFP use before revascularization, while again showing an increased rate of device-related complications, further underscoring the need for careful patient selection and continued device innovation.¹²

Technological innovation focuses on the development of next-generation percutaneous devices designed to provide higher flow rates and lower shear stress while operating with smaller percutaneous access profiles.

This review summarizes the current and evolving landscape of tMCS devices in CS. In addition, it discusses their role in HR-PCI, a clinical setting that frequently serves as the initial evaluation platform for next-generation tMCS technologies.

Existing Temporary Mechanical Circulatory Support Devices

Intraaortic Balloon Pump

The intraaortic balloon pump (IABP) is still the most widely used tMCS device worldwide. It operates by inflating a balloon placed in the descending aorta during diastole and deflating during systole, thereby augmenting coronary perfusion and reducing LV afterload. However, it is a passive device showing no effect on cardiac output or benefit with respect to clinical outcomes when tested in a randomized fashion.^{6,13–16} As a result, American and European guidelines do not recommend routine IABP use in AMI-CS.^{17–19} Corresponding adaptations for patients with heart failure–related CS are also expected in light of emerging evidence.^{16,19}

Extracorporeal Circulatory Support

ECLS in the venoarterial VA-ECMO provides the most comprehensive hemodynamic support among currently available systems (up to 7 L/min), offering both biventricular support and respiratory support using a venous outflow and arterial inflow cannula with an extracorporeal pump and oxygenator in between. However, important limitations remain, including the requirement for large-bore percutaneous femoral or surgical central access. In addition, an increase in LV afterload due to the retrograde return of blood into the descending aorta has been consistently discussed as a potential drawback, which may necessitate additional unloading strategies in some patients.¹ More recent in vivo studies suggest that this effect may have been overestimated, although randomized data are still lacking.²⁰ Despite the theoretically compelling support provided by ECLS, no RCT has demonstrated a benefit of its routine use in CS so far.^{5,8,21} Accordingly, American guidelines have recently issued a class III recommendation against its routine use in AMI-CS.¹⁸ However, in the case of complete circulatory collapse, it remains the only device that can be used.

Peripheral ECLS may increase LV afterload by raising mean arterial pressure and reversing flow in the aorta.²⁰

If the LV cannot accommodate the increased afterload, this will potentially result in elevated LV enddiastolic pressure, LV distension, pulmonary edema, ventricular thrombus, and impaired myocardial recovery.

Multiple methods are used to counterbalance this afterload increase, such as a reduction in right ventricular preload, resulting in a hemodynamic neutral effect. Venting refers to passive strategies that reduce left atrial or ventricular pressures to alleviate pulmonary congestion (eg, atrial septostomy, surgical or percutaneous left atrial vents). Left atrial VA-ECMO uses a fenestrated cannula to drain both atria. Although left atrial VA-ECMO may indirectly enhance biventricular decompression, it has not shown a mortality advantage over conventional configurations.²²

Unloading involves active reduction of LV pressure and volume, directly reducing wall stress and myocardial oxygen demand, thereby potentially promoting myocardial recovery, such as mAFP or IABP.^{23,24} This strategy with IABP is currently under investigation in the EULODIA randomized trial (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT07027202). MAFP unloading during VA-ECMO has been associated with improved outcomes in retrospective studies.²⁵ This strategy is under investigation in the UNLOAD-ECMO trial (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05577195).

In general, the need and kind of unloading or venting strategies need to be titrated according to patient clinical characteristics, ventricular phenotype, and hemodynamic features.^{23,24}

Left-Atrial to Femoral-Arterial Bypass Systems

The TandemHeart system was another percutaneous tMCS platform using an extracorporeal centrifugal pump to withdraw oxygenated blood from the left atrium via a transseptal cannula and return it to the arterial system. The device could generate ≈ 4 L/min of continuous flow while reducing LV preload and filling pressures.^{26,27} Despite these advantages, the need for transseptal puncture and large-bore cannulas has limited its widespread use, and together with limited evidence from RCTs led to a recent removal of this device from the market.

mAFPs

LV mAFPs, currently most prominently the Impella platform (J&J MedTech, Danvers, MA), are catheter-based support devices that provide active forward flow by directly transporting blood from the LV into the ascending aorta via an impeller mounted on a catheter shaft rotating at high speeds. In contrast to passive support systems, they augment cardiac output while simultaneously unloading the LV, thereby reducing LV enddiastolic pressure, wall stress, and myocardial oxygen demand. Different models provide varying levels of circulatory support. Currently, the percutaneously implanted Impella CP

(flow up to ≈ 3.7 – 4.3 L/min) and the surgically implanted Impella 5.5 flow (up to ≈ 5.5 L/min) are used for HR-PCI and CS.

The Impella CP is the only device in CS in which routine use has so far demonstrated a mortality benefit.³ This finding was observed in a highly selected population of patients with ST-elevation myocardial infarction-related CS who had a low risk of severe neurological injury, an LV ejection fraction $<45\%$, and no significant right ventricular dysfunction. Based on these results, American guideline recommendations for the routine use of mAFP in AMI-CS have been upgraded to a class IIa, level of evidence B recommendation.¹⁸

Of note, there is also a dedicated right-sided mAFP (Impella RP) available for the treatment of acute right ventricular failure, which is positioned across the right heart, thereby providing up to ≈ 4 L/min of flow. However, as isolated right ventricular failure in CS is rare, clinical experience remains limited, and randomized evidence is lacking.

A practical advantage, when compared with other MCS systems, is the ability to perform PCI using the same access sheath as is used for mAFP insertion. This reduces the number of access sites and subsequently the risk for potential vascular complications.²⁸

A further possible benefit of using mAFPs is the capability to provide information regarding hemodynamic variables such as intracardiac filling pressures if the device design allows pressure tracing.

Complications of Percutaneous Mechanical Circulatory Support

As outlined above, currently available tMCS devices remain associated with a considerable rate of serious complications.

Access-Related Injury and Bleeding

Vascular access complications are among the most common adverse events and include bleeding, pseudoaneurysm formation, and acute limb ischemia (ALI) related to the large-bore arterial access required for most devices.

ALI is associated with the vessel-to-sheath ratio—defined as the relationship between the inner diameter of the access vessel and the outer diameter of the implanted sheath—which is the primary determinant of residual perfusion. Femoral artery diameter varies considerably, with a mean of ≈ 7 to 8 mm in adults, though values <5 mm are not uncommon in women, the elderly, and those with peripheral arterial disease. Contemporary and next-generation tMCS devices impose substantial demands on this anatomy, as shown in Figure 1 with the required sheath sizes. Given that 1F equals ≈ 0.33 mm in diameter, a 14F sheath carries an outer diameter of roughly 4.7 mm—potentially occupying $>60\%$ of the lumen in a small-caliber femoral artery. In general, a

vessel-to-sheath ratio leaving less than ~30% to 40% of the luminal cross-sectional area patent has been associated with significantly elevated rates of ALI.²⁹

In major RCTs, rates of significant bleeding have ranged around 23% in the tMCS arms compared with 11% in the control arms.^{3,5,30} The risk of bleeding is further amplified by the profound alterations in coagulation associated with CS itself, as well as by the need for systemic anticoagulation due to the large foreign surface area of the devices. Importantly, bleeding complications in CS are associated not only with increased morbidity but also with higher mortality.³¹ The high shear forces generated by devices, such as ECMO and mAFPs, may also lead to proteolysis of high-molecular-weight von Willebrand Factor and a subsequent occurrence of acquired von Willebrand syndrome, another factor disposing patients to higher bleeding risk.³² Bleeding events are not always related to active use of tMCS, as a recent substudy of DanGer Shock reported a relative percentage of 50% of total bleeding events occurring after removal of the MCS device.³⁰ However, a substudy of the ECLS-SHOCK trial revealed no significant causal correlation of ECLS-related complications to 30-day mortality.³³ Further investigation is needed to better understand the complex interplay between bleeding risk, access site complications, and clinical outcomes.

Hemolysis

Pump-based support systems may induce hemolysis due to shear forces generated within the device. This

complication appears to be particularly pronounced with currently available mAFPs, where especially high pump rotational speeds and suction events may contribute to mechanical erythrocyte damage. Clinically relevant hemolysis may necessitate premature device removal and has been suggested as a potential contributor to the high rates of acute kidney injury requiring renal replacement therapy (42%) observed in the mAFP arm of DanGer Shock.³⁴ Major predictors of acute kidney injury specific to mAFP were suction events, higher pump speed, and longer duration of support.³⁴

Infections

Infectious complications may occur, particularly during prolonged device support. Large-bore vascular access sites and the presence of foreign material increase the risk of local and bloodstream infections, which may further aggravate organ dysfunction in critically ill patients.

Thromboembolism

Device-related flow disturbances, blood stasis, and the large foreign surface area of current systems may predispose to thrombus formation and subsequent thromboembolic events, including stroke.

Taken together, these limitations highlight the need for further technological innovation aimed at reducing vascular trauma, minimizing shear stress, and improving the hemocompatibility and safety profile of next-generation tMCS devices.

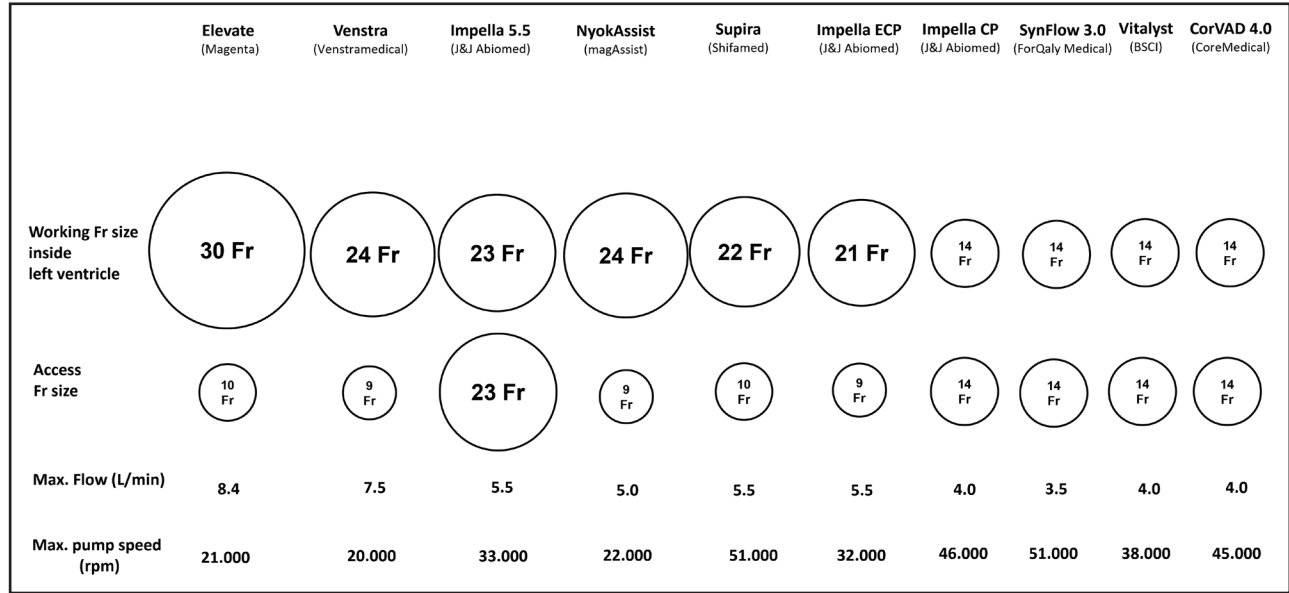


Figure 1. Current and next-generation microaxial flow pumps and the relation of working site, access size, maximum flow, and pump speed. Shown is the relation of working French (Fr) size in the left ventricle, access sheath Fr size, maximum flow (L/min), and maximum pump speed (revolutions per minute [rpm]) for currently available and emerging temporary active mechanical circulatory support devices. For more technical details and physics, see the detailed text. BSCI indicates Boston Scientific; CP, cardiac power; and ECP, expandable cardiac power.

Emerging and Next-Generation Percutaneous Mechanical Circulatory Support Devices

Requirements

Rapid advances in engineering and materials science have led to the development of next-generation tMCS devices aimed at improving hemodynamic support while reducing complications. These innovations focus on 3 key design principles: (1) smaller vascular access size; (2) higher achievable blood flow approaching normal cardiac output; (3) improved hemocompatibility with less hemolysis; and (4) less thrombosis risk with less requirement for anticoagulation.

Physics and Challenges

In tMCS devices designed to assist the failing heart, a key engineering challenge arises from the physical relationship between pump flow, rotational speed (revolutions per minute [RPM]), and the radius of the impeller or cannula. Under the simplified assumption that flow Q is proportional to the product of RPM and the square of the pump radius, the radius becomes a key determinant of achievable blood flow. Even modest increases in radius can produce a disproportionate gain in flow capacity. For example, a 20% increase in radius theoretically increases potential flow by $\approx 44\%$ at the same RPM. In the context of supporting cardiac output in patients with CS or during HR-PCI, this presents a design dilemma: effective hemodynamic support requires several liters per minute of additional flow to maintain systemic perfusion, yet vascular access limits the diameter of percutaneous devices that can be inserted through femoral or axillary arteries. Moreover, large-bore cannulas or introducer sheaths are associated with an increased risk of major access-site complications. Simply compensating for smaller device profiles by increasing RPM is not an ideal solution, as extremely high RPMs increase shear stress, thereby raising the risk of thrombosis and hemolysis with subsequent increased risk for renal replacement therapy. Also, RPM is limited by heat generation. Consequently, engineers must balance the competing demands of maximizing flow through larger radii while keeping device profiles small enough for safe vascular insertion.

Device performance depends not only on the relationship between pump radius and rotational speed. Although larger pump diameters generally enable higher flow generation at a given RPM, actual device performance depends on multiple additional factors, including pump design (axial versus centrifugal flow), pressure gradients, blood viscosity, inlet and outlet geometry, and overall system architecture. These relationships are commonly described by pressure-flow (H - Q) curves. Compared with centrifugal pumps used in VA-ECMO systems, axial flow pumps generally exhibit greater afterload sensitivity, such that increases in systemic arterial pressure may result in more pronounced reductions in pump flow. In addition, inlet and outlet flow conditions are critical determinants

of both performance and hemocompatibility. Excessive inlet velocities, inadequate ventricular filling, or device malposition may result in suction events or cavitation, potentially leading to reduced support, hemolysis, and device instability.

Different Design mAFPs

The further development of tMCS technologies currently focuses predominantly on mAFPs. Owing to their relatively straightforward implantation and single-site arterial access, this design remains particularly attractive in clinical practice. In general, mAFP devices can be subdivided into expandable and nonexpandable devices. Currently, available systems, such as the Impella CP and Impella 5.5, rely on nonexpandable designs, while newer devices either refine this concept or incorporate expandable technologies to enable higher flow with smaller access profiles.

Nonexpandable Devices

Vitalyst

The VITALYST device (Boston Scientific, Marlborough, MA) is a mAFP that aids forward blood flow of up to 4 L/min. Key design elements include a magnetically driven impeller that eliminates the need for a purge solution for anticoagulation, reduces hemolysis, and enables a smaller drive shaft profile to lower the risk of vascular complications. In addition, it has an optimized catheter with additional bends and motor design to improve positional stability (Figures 1 and 2A).

Technical success and safety could be demonstrated in an early feasibility study analyzing an HR-PCI population in 15 patients.³⁵ The first RCT in HR-PCI is planned to start enrollment in 2026 (Table).

CorVAD 4.0

The CorVAD 4.0 (CoreMedical, HXCORMED, Shenzhen, China) is a mAFP that allows a maximum flow rate of 4 L/min, using a comparably large access sheath size of 14F (Table; Figures 1 and 2B).

A prospective multicenter single-arm study ($n=118$) in China showing the device's safety and efficacy in HR-PCI has already been completed in 2024, including a total population of 118 patients with a follow-up duration of 30 days (Table).

The REC-CHIPMCS trial (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06373120), an RCT comparing the CorVAD 4.0 device against VA-ECMO in a population ($n=260$) undergoing HR-PCI, was presented recently, showing a statistically significant reduction of absolute risk by almost 15% in favor of the CorVAD 4.0 device regarding a composite end point of death, stroke, myocardial infarction, and procedure related adverse events including bleeding at 30 days ($P=0.003$). This was primarily driven by a lower rate of bleeding events (BARC, 3–5) with a rate of 5.3% in the mAFP group versus 11.8% in the VA-ECMO arm.

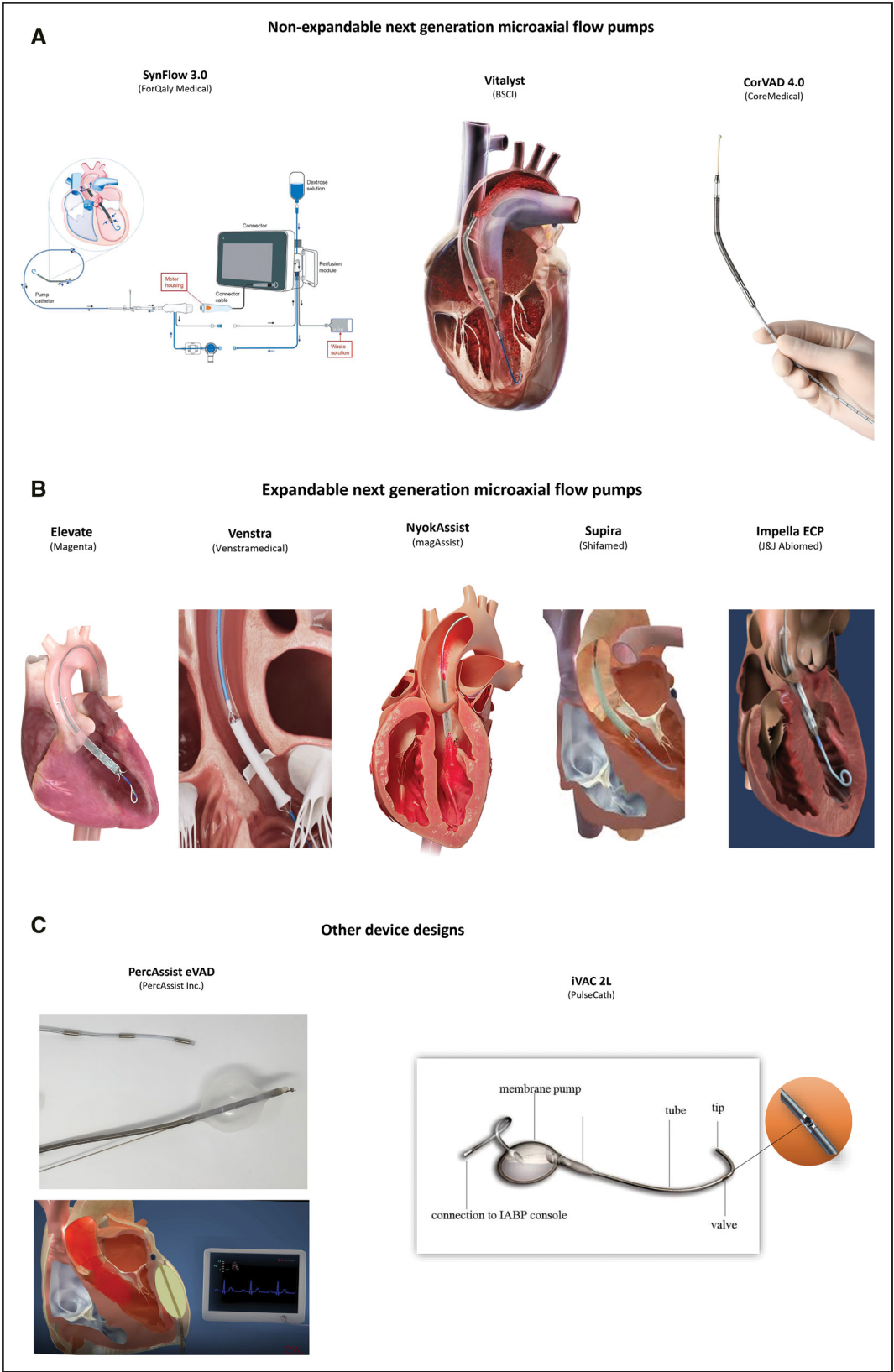


Figure 2. Next generation mechanical circulatory support devices.
A, Next generation nonexpandable microaxial flow pump devices under development. Shown are 3 different nonexpandable microaxial flow pump devices. **Left**: Device including controller and connection systems plus purge solution. **Middle**: Device without purge placed in the left ventricle. **Right**: device outside the body. For more technical details, including pump flow, access sheath size, and maximum pump speed, see Figure 1. **B**, Next generation expandable temporary mechanical circulatory support (tMCS) devices under development. (*Continued*)

In a similar design to the above-described DanGer Shock trial, which showed superiority of Impella CP in selected ST-elevation myocardial infarction-CS, the CorVAD 4.0 device is also planned to be tested in a cohort of AMI-CS patients. The not-yet-recruiting CorVAD AMICS trial (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06886113) plans to enroll a total of 269 patients with AMI-CS, excluding those with out-of-hospital cardiac arrest. Patients will be randomized to either CorVAD 4.0 implantation before PCI or standard of care in a 1:1 ratio (Table).

Another currently enrolling RCT (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06646224) is investigating the device's potential benefit in a planned cohort of $n=50$ patients with CS caused by cardiomyopathy versus a control group without routine mechanical circulatory support. The study was started in late 2024, is currently enrolling at a single center in China, and represents one of the few ongoing studies investigating a contemporary mAFP in heart failure-related CS (Table).

SynFlow 3.0

The SynFlow 3.0 device is a mAFP that also stems from China (ForQaly Medical, Shanghai, China). Its design resembles the Impella CP device, with a low-profile vascular access sheath of 9F and a flow rate of ≈ 3.1 L/min (Figures 1 and 2A).

This device has been tested in an early feasibility and safety HR-PCI study including 10 patients at a single center in China (Table). Results showed technical success in the absence of serious adverse events, while there is a lack of clear documentation regarding bleeding events in the published data.³⁶ Furthermore, it has also been tested in an RCT (PERSIST III; URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06853470) with randomization against VA-ECMO in HR-PCI, with no significant difference between the 2 treatment arms regarding a composite end point consisting of all-cause mortality, BARC 3 to 5 bleeding, acute kidney injury, cardiopulmonary resuscitation, cardiac/surgery, or device-related limb ischemia. Detailed information on occurrence rates of the individual components is not available so far, which may conceal differences, for example, bleeding events (Table).

KARDION

Another device presently undergoing active investigation is the KARDION Cory P4 device (Kardion GmbH, Stuttgart, Germany), which is currently tested in an RCT

(PICANTE; URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06445608) including patients with HR-PCI, where included patients are randomized to either the company's novel device or a commercially available mAFP. Other than being a mAFP, which is delivered over the aortic valve, no technical details are currently disclosed.

Expandable mAFPs

Expandable microaxial pumps may represent a major step forward in tMCS technology. Unlike earlier devices that require large arterial sheaths, these pumps can be delivered through small catheters and then expand to full operating diameter once inside the body. To achieve this goal, most systems under development use self-expanding structures, commonly based on nitinol frameworks, that can be delivered through a low-profile catheter and subsequently expand to a substantially larger functional diameter once positioned across the aortic valve. By increasing the effective pump diameter after deployment, these devices may achieve similar flow rates at lower rotational speeds. This may theoretically reduce erythrocyte exposure to high-shear environments and improve hemocompatibility. However, the relationship between rotational speed, shear stress, and blood trauma is complex and depends on multiple device-specific factors, including impeller geometry, local flow patterns, turbulence generation, recirculation zones, inlet and outlet flow characteristics, and erythrocyte exposure time to regions of elevated shear stress. In addition, these potential advantages must be balanced against engineering challenges, such as reliable deployment and retrieval, maintenance of structural integrity during prolonged support, and ensuring consistent expansion under varying anatomic conditions. Whether these theoretical advantages translate into clinically meaningful reductions in hemolysis, thrombosis, or vascular complications remains to be demonstrated in prospective clinical studies.

Impella ECP

The most recent addition to the Impella line is the Impella ECP (J&J, Abiomed, Danvers, MA). This device is estimated to provide a mean flow of 3.8 L/min with the ability to enhance its hemodynamic support to a maximum flow capacity of 5.5 L/min. The system is deployed via a 9F vascular access sheath (Figures 1 and 2B).

The Impella ECP could already show safety and efficacy in a pivotal HR-PCI trial (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05334784)

Figure 2 Continued. Shown are 5 different expandable microaxial flow pump devices with the different expanding systems in place in the left ventricle. For more technical details, including pump flow, access sheath size, and maximum pump speed, see Figure 1. **C**, Other tMCS device types. **Left upper:** Direct cardiac compression device with balloon. **Left lower:** Direct cardiac compression device with balloon in place in the pericardial space secured anterior to the left ventricle. Balloon pneumatically air-driven with systolic inflation synchronized to the QRS complex. **Right:** Pulsatile extracorporeal support system driven by a conventional intraaortic balloon pump console. The 17F flow catheter is across the aortic valve with the distal tip located in the left ventricle. The outflow valve is shown on the right, magnified. BSCI indicates Boston Scientific; and ECP, expandable cardiac power.

Table. Study Status of Emerging Percutaneous Mechanical Circulatory Support Technologies

	Type of study	Sample size	Current status	Publication status	Outcome (if completed)
VITALYST	HR-PCI FIH	5	Completed	...	Technical and clinical success
	HR-PCI multicenter EFS (NCT06132568)	15	Completed	Croce et al <i>Circ Cardiovasc Interv</i> 2026;epub ³⁵	Safety and early feasibility at 30-day follow-up
	HR-PCI RCT (VITAL-PCI; NCT07223788)	≈400 planned	Planned	...	
CorVAD 4.0	HR-PCI EFS (NCT06198998)	118	Completed	...	Safety and early feasibility at 30-day follow-up
	HR-PCI RCT: CorVAD 4.0 vs VA-ECMO (REC-CHIPMCS; NCT06373120)	260	Completed	Abstract TCT 2026	ARR 15% in favor of CorVAD 4.0, fewer bleeding events in the mAFP arm (5.3% vs 11.8%)
	HF-CS RCT: CorVAD 4.0 vs SoC (NCT06646224)	50 planned	Recruiting	...	
	AMI-CS RCT: CorVAD 4.0 vs SoC (CorVAD AMICS; NCT06886113)	269 planned	Not yet recruiting	...	
SynFlow 3.0	HR-PCI EFS	10	Completed	Wang et al ³⁶	Safety and early feasibility at 30-day follow-up
	HR-PCI RCT: SynFlow 3.0 vs VA-ECMO (PERSIST III; NCT06853470)	222	Completed	Abstract TCT 2025	No difference in MACE events at 30-days, no detailed information regarding bleeding events
KARDION Cory P4	HR-PCI RCT: KARDION Cory P4 vs Commercial Control mAFP (PICANTE; NCT06445608)	310 planned	Recruiting	...	
Impella ECP	HR-PCI pivotal trial with continued access (NCT05334784)	256 in pivotal trial; up to 600 with continued access protocol	Pivotal trial completed	Abstract TCT 2024	30-day MACCE rate 6,3% (n=15)
Supira pLVAD	HR-PCI OUS FIH	6	Completed	Abstract SCAI 2023	Technical and clinical success
	HR-PCI OUS EFS (NCT05864248)	71	Completed	...	Technical and clinical success
	HR-PCI US EFS (SUPPORT I; NCT06087575)	15	Completed	Kandzari et al ³⁷	Safety and early feasibility at 30- and 90-day follow-up
	HR-PCI US pivotal study (SUPPORT II; NCT07296744)	358	Planned	...	
	HR-PCI with prolonged support EFS (NCT07407816)	45 planned	Recruiting	...	
	CS pivotal study	Unknown	Planned	...	
Elevate	HR-PCI OUS FIH (ELEVATE I; NCT06099548)	20	Completed	...	Safety and early feasibility at 30-day follow-up
	HR-PCI US EFS (ELEVATE II; NCT05727059)	18	Completed	...	Safety and early feasibility at 30-day follow-up
	HR-PCI pivotal RCT: Elevate vs Impella (ELEVATE III; NCT07001332)	290	Recruiting	...	
	AMI and HF-CS FIH (ELEVATE IV; NCT07293923)	10 planned	Recruiting	...	
Venstra	HR-PCI FIH	Unknown	Planned	...	
NyokAssist	HR-PCI single case as part of large animal EFS	1	Completed	...	Technical success, no adverse events
	HR-PCI RCT: NyokAssist vs IABP (SAVE-PCI; NCT07010965)	236	Completed	Abstract TCT 2025	Enrollment completed, 30-day MACCE rates similar, follow-up ongoing
Percassist eVAD	FIH Trial in CS (SCAI E)	7	Completed	ASAIO Abstract 2024	Technical and clinical success
	Larger Trial in CS (SCAI C-D)	25 planned	Planned	...	

(Continued)

Table. Continued

	Type of study	Sample size	Current status	Publication status	Outcome (if completed)
iVAC 2L	HR-PCI FIH	14	Completed	den Uil et al ³⁸	Technical success, no relevant adverse events
	HR-PCI FIH (PULSE study)	29	Completed	Bastos et al ³⁹	Technical success, hemodynamic effects
	HR-PCI registry	293	Completed	Bulum et al ⁴⁰	Technical success, no relevant adverse events
	RCT HR-PCI (UNLOAD CHIP) (NCT07380217)	98	Planned		
	RCT HR-PCI (PULSE II RCT; (n.a.)	368	Planned		
	HR-PCI RCT in China (n.a.)	316	Planned	Described in Bulum et al ⁴⁰	

AMICS indicates acute myocardial infarction cardiogenic shock; ARR, adjusted relative risk; CS, cardiogenic shock; EFS, early feasibility study; FIH, first in human; HR-PCI, high-risk percutaneous coronary intervention; MACCE, major adverse cardiocerebrovascular events; mAFP, microaxial flow pump; n.a., not available; OUS, out of the United States; PULSE, Pulsecath Mechanical Support Evaluation; RCT, randomized controlled trial; SCAI, Society for Cardiovascular Angiography and Interventions; SUPPORT, Early Feasibility Study of the Supira System in Patients Undergoing High-Risk Percutaneous Coronary Intervention; TCT, transcatheter cardiovascular therapeutics; UNLOAD CHIP, Use of Mechanical Left Ventricular Unloading in Complex Higher-Risk Indicated Procedures; and VA-ECMO, venoarterial extracorporeal membrane oxygenation.

encompassing 256 patients, with a major adverse cardiac and cerebrovascular event rate of 6.3% (n=15) at 30 days, markedly below the previously set performance goal of 24.4% (Table). Although the primary inclusion target has been reached, the trial is still including patients as part of a continued access protocol. The device is currently under submission for approval by the United States Federal Device Administration (FDA).

Supira pLVAD (Supira Medical)

The Supira pLVAD (Shifamed, Campbell, CA) is an expandable mAFP that can provide up to 5.5 L/min maximum flow capacity and is inserted via a 10F sheath (Figures 1 and 2B).

First-in-human and early feasibility studies outside the United States and in the United States in HR-PCI cohorts (SUPPORT I [Early Feasibility Study of the Supira System in Patients Undergoing High-Risk Percutaneous Coronary Intervention]; URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06087575) have recently been published, demonstrating technical success and safety (Table).³⁷ Beyond its performance in the early feasibility setting, it is also currently being investigated as a single-arm study in a population with a requirement of hemodynamic support beyond the initial HR-PCI procedure (SUPPORT II; URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT07407816). Furthermore, a study in AMI-CS is planned to start in 2026 (Table).

Elevate

The Elevate System (Magenta Medical Ltd, Kadima, Israel) is centered on a self-expanding axial-flow pump engineered to deliver high-output hemodynamic support via a 10F transfemoral access. The catheter-mounted pump is introduced in a crimped configuration within a 10F outer-diameter delivery catheter. On positioning within the LV, the device is unsheathed to a functional

diameter of 30F. The pump is driven by a sterile external motor unit connected to a dedicated bedside controller (Figures 1 and 2B).

The system operates across 10 speed settings, achieving rotational speeds up to 21 000 RPM. At a pressure gradient of 60 mmHg, mean flow exceeds 5.4 L/min, with peak systolic flows surpassing 8.4 L/min, enabling sufficient support to fully replace native cardiac output.

In the first-in-human ELEVATE I study (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT06099548) conducted in Georgia, 20 patients undergoing HR-PCI were analyzed (Table). Outcomes from the initial 14 patients, all with left main coronary artery disease, have been previously reported. The subsequent US-based early feasibility study ELEVATE II (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05727059) enrolled 18 patients undergoing HR-PCI, showing sufficient safety. The ongoing pivotal RCT ELEVATE III (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT07001332) compares the Elevate system with the Impella CP in HR-PCI and was designed to form the basis for future regulatory approval of the system in the United States. Furthermore, the clinical feasibility study ELEVATE III (URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT07001332) currently evaluates the device's performance in patients with CS due to both AMI and HF (n=10) in centers in Georgia and Israel (Table).

Venstra

The Venstra device (Venstramedical, Newcastle, Australia) is a mAFP that is characterized by a low-profile design, as the catheter sheath needed for vascular access measures only 9F, with the device expanding its diameter when placed across the aortic valve to 24F (Figures 1 and 2B). It is able to provide a flow of 5 to 7 L/min at a pressure of 60 to 80 mmHg, with the ability of

providing full support with a flow up to 10 L/min against 30 mm Hg of pressure.

After successful in vivo experiments showing the technical function of the device, a first-in-human trial in a small cohort of HR-PCI patients is currently on the horizon (Table).

NyokAssist

The NyokAssist is another mAFP currently under development by the Chinese company magAssist (magAssist Co Ltd, Suzhou, China). It is inserted via a 9F vascular access sheath and is supposed to provide a peak flow of >4.5 L/min (Figures 1 and 2B).

A recent abstract showcased data from an early feasibility study in a swine model. Although no patient-centered trial has been conducted yet, a first-in-human use in a case of HR-PCI is documented without any adverse events (Table).

Limitations of mAFP

Despite ongoing technological innovations in mAFP devices, with the prospect of improved hemodynamic support and reduced complication rates, important limitations remain. These devices are primarily restricted to LV support and therefore depend on adequate native right ventricular function, although combined right-sided support is theoretically possible. In addition, mAFP systems do not provide respiratory support or gas exchange. Their use may also be limited by anatomic constraints, including vascular access limitations and structural pathologies, such as severe aortic regurgitation, the presence of mechanical aortic valve prostheses, or intracardiac thrombi. Another important aspect is the correct placement of the device, as malposition may lead to suction events, reduced flow, and device migration, all of which can compromise support and increase the risk of complications. Furthermore, ventricular size influences device performance: a small LV may predispose to suction and cavitation events, whereas an enlarged LV may increase the risk of device migration. Furthermore, the advantage or disadvantage of externally placed motors with long drivelines, the durability of these devices, and the performance of self-expandable systems with risk of severe aortic regurgitation in case of device malfunction when in place in the aortic valve requires additional investigations.

Other Devices

Percassist eVAD

Another interesting novel VAD device, which does not follow the technical design of mAFPs, is the Percassist eVAD (PercAssist Inc, CA), belonging to the family of direct cardiac compression devices.⁴¹ The device consists of a direct cardiac compression balloon positioned

within the pericardial space through a percutaneous sub-xiphoid approach and secured anterior to the LV using a 20F vascular sheath. The balloon is pneumatically driven with ambient air, with systolic inflation synchronized to the QRS complex by means of electrocardiographic sensing via a subcutaneous thoracic lead, while the pump and control console remain extracorporeal. More information on this device type has recently been summarized.⁴¹

A first-in-human study was already conducted in a small cohort (n=7) of patients with CS and SCAI Shock Classification E, where its safety and efficacy could be demonstrated. A subsequent trial including a larger population in SCAI stage C-D is currently under planning (Table; Figure 2C).

PulseCath iVAC 2 L

The PulseCath iVAC 2 L (PulseCath BV, Amsterdam, the Netherlands) is a pulsatile extracorporeal support system driven by a conventional IABP console. This console is connected to a 17F flow catheter across the aortic valve, with the distal end being located in the LV. The device is deployed via femoral access and requires an 18F vascular sheath. It aspirates blood from the LV during systole and ejects it into the ascending aorta during diastole; this adds a limited support of 1.5 L/min to the existing cardiac output and thus delivers a better hemodynamic performance than the traditional IABP.⁴² The device has gained the CE mark in Europe.

After an initial 14-patient study,³⁸ the PULSE study (PulseCath Mechanical Support Evaluation) included 29 patients undergoing HR-PCI while using extensive hemodynamic measurements and could demonstrate active LV unloading as well as reduced myocardial oxygen uptake, especially in patients with high filling pressures at baseline.³⁹ Data from an international multicenter registry demonstrated an improvement in hemodynamics with a low rate of complications.⁴⁰ The planned UNLOAD-CHIP trial (Use of Mechanical Left Ventricular Unloading in Complex Higher-Risk Indicated Procedures; URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT07380217) is designed to evaluate the composite end point of 30-day all-cause mortality or CS in 98 stable high-risk patients with impaired LV function randomized to iVAC2L or no MCS. The PULSE II trial will compare the incidence of major adverse events at 90 days between 368 elective HR-PCI patients with randomization to iVAC2L or Impella CP. Furthermore, a multicenter RCT in China will assess the incidence of 30-day major adverse events in 316 patients with stable HR-PCI randomized to iVAC2L or IABP support (Table; Figure 2C).⁴⁰

Despite CE mark approval, adoption in routine practice has been modest, which may be related to the large access site in relation to the only modest support. Furthermore, availability is currently restricted in several markets, and robust outcome data are lacking.

Regulatory Challenges

tMCS device approval faces substantial regulatory challenges when seeking approval in the United States through the FDA compared with obtaining the CE mark in Europe under the European Commission medical device regulatory framework. These devices are typically classified as high-risk (class III) technologies because they directly interact with the cardiovascular system and are used in critically ill patients with high mortality risk, such as those experiencing CS or undergoing HR-PCI. As a result, the FDA generally requires a rigorous premarket approval process supported by robust clinical evidence demonstrating safety and effectiveness, often including RCTs, extensive preclinical testing, and longer-term durability and longer-term follow-up. In addition, postapproval studies are frequently mandated to assess real-world performance in broader patient populations. A key challenge in this context is that pivotal trials in CS are difficult to conduct due to the emergent nature of the condition, heterogeneous patient presentations, and high baseline mortality rates approaching 50% despite modern treatment strategies. In addition, patients with CS often require several days of continuous support, which imposes additional demands on device performance and durability. Furthermore, the assessment of device-related complications in this population of severely critically ill patients remains highly complex. As a result, many novel tMCS devices are initially evaluated in the HR-PCI setting, where procedures are more controlled, patient populations are more stable, and trial execution is more feasible. In contrast, the European CE marking process historically allowed earlier market entry for tMCS devices by emphasizing conformity with safety and performance standards rather than requiring the same level of premarket clinical evidence demanded by the FDA. This difference enabled several devices to become available in Europe years before receiving FDA approval or expanded indications in the United States. For example, certain Impella systems obtained CE marking for use in HR-PCI and CS before similar indications were granted by the FDA through premarket approval expansions. Although the newer European Medical Device Regulation has increased clinical evidence requirements, significant structural differences remain between the 2 systems. The FDA review process is centralized and conducted by a single regulatory authority, whereas CE marking involves multiple notified bodies evaluating conformity assessments, which can create variability in interpretation and timelines. For device developers, navigating these regulatory pathways often requires separate clinical programs and documentation strategies tailored to each jurisdiction. Additionally, rapid technological iteration—common in catheter-based pumps and circulatory assist platforms—can trigger the need for new regulatory

submissions when design modifications affect device performance or safety. These challenges are amplified by the urgent clinical context in which tMCS devices are used, where clinicians seek rapid access to innovative hemodynamic support technologies, but regulators must ensure patient safety through stringent oversight. Consequently, balancing timely innovation with robust evidence generation remains a central regulatory challenge for tMCS.

In general, all trials should follow the Shock Academic Research Consortium end point definitions.⁷ In light of the high costs—sometimes also long-time for patient enrollment—with the conduct of randomized trials in CS, there are also opportunities for implementation of innovative trial designs such as win-ratio or Bayesian analyses enabling the enrollment of a lower number of patients than in classical trial designs.^{43,44} Furthermore, in particular in the United States, where exception from informed consent is required for CS trials,⁴⁵ trial augmentation methods, in which observational data are used to augment the precision and efficiency of outcome estimations in trials, are discussed. By harnessing the additional information provided by external data sources, these methods can reduce the sample size needed to adequately power future trials. This method is especially promising for a setting like CS in which randomized trials are especially challenging to conduct, and observational data sets are extensive.^{46,47}

Given the most recent results of the CHIP-BCIS 3 trial (Controlled Trial of High-Risk Coronary Intervention With Percutaneous Left Ventricular Unloading – British Cardiovascular Intervention Society 3) not showing a benefit of routine LV unloading with the Impella CP in HR-PCI compared with standard of care, with even higher cardiovascular mortality in the Impella arm, questions about the overall process of initial regulatory approval in HR-PCI settings.¹¹ It may also question the design of RCTs to show noninferiority of newer mAFP devices in comparison to existing FDA-approved devices in the HR-PCI setting when the overall indication for tMCS is not proven. This most likely will need a refinement of the regulatory approval process.

Future Directions

In addition to technical engineering advances with smaller access sites, more flow with physiological cardiac output flow rates, possible integration of continuous hemodynamic monitoring, reduction of hemolysis, and avoidance/minimization of anticoagulation, it will be important to define regulatory processes for approval particular in the CS setting. This must also include streamlining of regulatory processes without an excess in adverse events and risks for the patients.

From a clinical perspective, the indication when to use which device in which patient and which phenotype is of

utmost importance, and much more evidence is needed. Ongoing and upcoming RCTs will hopefully define the role of tMCS in different causes of CS and CS phenotypes, including broader patient populations of AMI-CS than included in the DanGer Shock trial, heart failure-related CS, and secondary CS.

Furthermore, more research is required for the combination of devices, including right heart support, and optimal timepoints for escalation to durable devices or heart transplantation.

CONCLUSIONS

tMCS devices have become an essential component of modern CS management, whereas the indication in the HR-PCI setting has recently been challenged.¹¹ Future trials are required to elucidate their role in this setting.⁴⁸ Although early passive devices, such as IABP, provide no relevant hemodynamic augmentation, newer active tMCS devices offer significantly greater circulatory support.

Rapid technological innovation is driving the development of next-generation devices capable of delivering near-physiological blood flow through more limited vascular access. These advances have the potential to expand the role of tMCS and improve outcomes in patients with CS. However, regulatory approval processes are complex and time-consuming, thus limiting the use of novel device technology. In light of a recent negative RCT in the HR-PCI setting also regulatory approval may need refinement.

Future research should focus on RCTs evaluating device selection, timing of support, and optimal patient populations to maximize the clinical benefits of these technologies.

ARTICLE INFORMATION

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